Saturated Superfluid Vacuum I

Topological Defects in a Saturated Superfluid Vacuum:
The Geometric Origin of Matter and
the α-Harmonic Mass Ladder

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Abstract

We propose that the physical vacuum is a compressible, inviscid superfluid near saturation density — a single postulate from which the structure of fundamental particles and constraints on the fine-structure constant emerge. Particles are stable topological defects: quantised toroidal vortex rings (leptons) and spherical breathers stabilised by trefoil vortex skeletons (baryons).

From the logarithmic Schrödinger Lagrangian we derive three results from first principles. First, the sound-speed condition $c_s = c$ fixes the stiffness parameter $b = m_0c^2/(2\rho_0)$ and the healing length $\xi = \hbar/(m_0c)$. Second, minimisation of the three-term vortex-ring energy functional over torus radius R and core radius a yields the unique equilibrium $R_e^* = \xi/\alpha$, from which the vacuum density ρ_0 is fixed via the Lamb formula with no free parameters beyond the two empirical inputs m_e and α. Third, the ratio of the chiral-shear speed $c_\perp$ to the sound speed c_s identifies $\alpha \approx 1/137$ as a geometric property of the medium.

These results establish the base energy scale $\mu_0 \equiv m_ec^2/\alpha \approx 70\,\text{MeV}$ as the energy of a minimal vortex filament segment of the electron Compton circumference. The charged pion, as the minimal vortex-antivortex bound state carrying two winding numbers, then has mass $m_{\pi^\pm} = 2\mu_0$, agreeing with CODATA at 0.34%.

Direct 3D toroidal Bogoliubov–de Gennes projection with chiral-shear coupling is developed as a diagnostic for the muon rung, but the present scalar operator does not close the identification. The reduced helicity-basis hit near $(3/2)\mu_0$ is not Galerkin-robust, a direct $m_\varphi = \pm 1$ grid-sector solve finds no corresponding low branch, and a Volovik-style Berry-phase audit finds trivial holonomy for the implemented $\mathcal{L}_\perp$ block. The muon therefore remains a candidate empirical rung rather than a derived eigenfrequency. For the proton, the empirical scaling $m_pc^2 = N_Y \cdot F \cdot \mu_0$ with $N_Y \approx 3.007$ and form factor $F \approx 4.4$–4.47 at the inter-strand half-spacing cutoff $R \approx 1.18\,\xi$ yields $m_p \approx 927$–$941\,\text{MeV}$ (1.2%–0.3% from CODATA, depending on the cutoff convention). The neutron is given a structural derivation as the surface-locked composite $\mathcal{N} = \mathcal{T} \cup_{\Sigma_p} \mathcal{E}$ of the proton trefoil and an electron torus compressed onto the breather skin: charge, spin, spin-statistics composition, and baryon/lepton number follow without fitting from boundary-framed vs bulk-framed defect classification, and the mass splitting decomposes as $\Delta E = m_ec^2 + \Delta E_{\text{surf}}$ with ΔE_{surf} identified as the β-decay endpoint $0.782\,\text{MeV}$; quantitative pinning of ΔE_{surf} inherits the proton-closure work and is tracked under issue #30.

Two empirical inputs (m_e, α) are used. The remaining predictions are consistent with the observed lepton and light-hadron spectrum at the $\lesssim 1.5\%$ level under the present effective theory, with explicit dependence on the regularisation cutoff R and the meridional domain size L_{box} documented in the body. A dedicated Claim Status section (§8) tags every quantitative claim as *derived*, *structural*, *candidate*, or *ansatz*, and records the issue #30 closure ledger for the candidate-grade branches. The remaining open problems for future work are: a changed microscopic ladder ingredient if the muon rung is to be derived, topology-preserving static infrastructure for the proton branch, and a closure-grade dynamic reconnection simulator for Paper II. The previously headlined 0.3% proton-mass agreement is the precision at a specific calibration cutoff $R \approx 1.18\,\xi$ and is reframed in this paper as a $\sim$1.5% match with documented cutoff sensitivity.

Contents

1 Introduction: The Crisis of Abstraction

The Standard Model predicts particle masses, charges and interaction rates with extraordinary precision, yet it provides no mechanism for *why* these quantities take their observed values. Some 19 free parameters—particle masses, coupling constants, mixing angles—must be inserted by hand. At the same time, attempts to quantize gravity perturbatively produce non-renormalizable divergences, and roughly 95% of the universe's energy content must be assigned to unknown dark components inferred only from gravitational anomalies.

We propose that this impasse reflects a missing ontological layer. The vacuum is not empty: it is a physical substance—a saturated superfluid. All phenomena emerge from its hydrodynamics. This idea has deep roots. Volovik has shown that the low-energy physics of superfluid ^{3}He reproduces the structure of the Standard Model, with Lorentz symmetry, gauge bosons, and chiral fermions emerging as collective excitations [1]. Barceló, Liberati, and Visser established the analogue-gravity programme: acoustic perturbations of any irrotational, barotropic fluid obey a curved-spacetime wave equation, so gravitational phenomenology emerges from fluid dynamics [3]. Hu has argued that spacetime itself is an emergent concept from a more fundamental substrate [4].

The present work pursues a specific realisation of this programme: a logarithmic nonlinear Schrödinger (LogSE) field theory for the vacuum order parameter, supplemented by a chiral-shear term. Stable topological defects of this field—toroidal vortex rings and spherical breathers—are identified with fundamental particles. The mass spectrum of these defects organises into a harmonic ladder governed by the fine-structure constant $\alpha \approx 1/137$.

This paper is organised as follows. Section 2 states the fundamental postulate, derives the equations of motion from the LogSE Lagrangian, derives the sound speed, and establishes the origin of α from the chiral-shear coupling. Section 3 establishes the equilibrium vortex geometry and the vacuum density from the Lamb vortex-ring formula, taking m_e and α as empirical inputs, and presents the proton and neutron models. Section 4 develops the α-harmonic mass ladder, including the pion (analytically), the muon (dynamically motivated bracket), and the proton (numerically extracted, cutoff-dependent). Section 5 sketches the emergence of forces. Sections 6–7 give the outlook and conclusions. Appendices contain the numerical minimisation and the BdG eigenvalue equations together with their numerical solution.

2 The Fundamental Postulate

The physical vacuum is a continuous, compressible, inviscid superfluid at near-saturation density ρ_0, with background pressure $P_0 \gg \rho_0 c^2$. All known physics emerges from the hydrodynamics of this single medium; no additional fields or symmetry-breaking mechanisms are required at the foundational level.

The defining characteristics are:

- **Compressibility and high stiffness.** The equation of state is such that small density perturbations propagate as pressure waves at speed $c_s = \sqrt{\partial P/\partial \rho}$. At the background state, we require $c_s = c$ (the observed speed of light) to be constant (Lorentz-invariant in the low-momentum limit). The condition $P_0 \gg \rho_0 c^2$ ensures the medium remains far from the laminar-to-turbulent transition under ordinary stresses, while still allowing topological defects to form under extreme local strain.

- **Inviscid flow.** Zero (or exponentially suppressed) viscosity guarantees dissipation-free long-range coherence—persistent currents, quantized circulation, and stable solitons. This is the hydrodynamic analogue of unitarity and conservation laws.

- **Light as sound.** Electromagnetic waves (and massless modes generally) are long-wavelength longitudinal/transverse pressure oscillations in the superfluid. Thus,

$$c = c_s(\rho_0) = \sqrt{\left.\frac{\partial P}{\partial \rho}\right|_{\rho_0}}. \tag{1}$$

  In logarithmic superfluid models (common in SVT literature), the pressure-density relation takes the form $P \propto -\rho \ln(\rho/\bar{\rho})$ [1, 2], yielding constant c_s at low excitations while allowing deformed dispersion at high momenta.

- **Special relativity as emergent hydrodynamics.** Objects composed of stable topological defects (vortices/breathers) experience length contraction and time dilation as equilibrium responses to motion through the medium. A moving defect induces a wake; to minimise energy/disruption the defect contracts in the propagation direction (Lorentz–FitzGerald style). The medium itself remains undetectable in Michelson–Morley-type interferometers because the "ether wind" is canceled by the defect's own flow field—the drag is internal to the object-medium system.

This postulate recovers special relativity in the low-velocity, low-excitation limit without assuming a priori Lorentz symmetry [1, 3]; the symmetry emerges from the constancy of c_s and the irrotational nature of small perturbations (potential flow). At higher energies/momenta, deviations from exact Lorentz invariance are expected (deformed dispersion relations), consistent with hints from ultra-high-energy cosmic rays and some quantum-gravity phenomenology [3, 10].

2.1 Effective Lagrangian and Equations of Motion

The dynamics of the saturated superfluid vacuum are governed by an effective field theory for the complex order-parameter field Ψ (normalised so $|\Psi|^2 = \rho/\rho_0$). Following the superfluid-vacuum framework of Volovik [1], we adopt a logarithmic nonlinear Schrödinger Lagrangian:

$$\mathcal{L} = \frac{i\hbar}{2}\big(\Psi^* \partial_t \Psi - \Psi\, \partial_t \Psi^*\big) - \frac{\hbar^2}{2m_0}|\nabla\Psi|^2 - V(\rho), \tag{2}$$

where the potential is

$$V(\rho) = -b\,\rho\big[\ln(\rho/\bar{\rho}) - 1\big] + V_0, \tag{3}$$

with $b > 0$ a stiffness parameter and $\bar{\rho}$ the saturation density. Applying the Euler–Lagrange equation to (2), using $\partial V/\partial \Psi^* = (dV/d\rho)\rho_0 \Psi$ and $dV/d\rho = -b\ln(\rho/\bar{\rho})$, yields the logarithmic Schrödinger equation (LogSE):

$$i\hbar\, \partial_t \Psi = -\frac{\hbar^2}{2m_0}\nabla^2\Psi - b\rho_0 \ln\left(\frac{\rho_0|\Psi|^2}{\bar{\rho}}\right)\Psi. \tag{4}$$

Madelung representation. Writing $\Psi = \sqrt{\rho/\rho_0}\, e^{i\theta}$ with superfluid velocity $\mathbf{v} = (\hbar/m_0)\nabla\theta$, the LogSE (4) separates into real and imaginary parts. The imaginary part gives the exact continuity equation

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{5}$$

The real part, after dividing by ρ, gives an Euler equation

$$\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\frac{1}{m_0 \rho}\nabla P(\rho) + \frac{\hbar^2}{2m_0^2}\nabla\left(\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}\right), \tag{6}$$

where the pressure follows the logarithmic equation of state $P(\rho) = b\rho\ln(\rho/\bar{\rho})$. The last term is the quantum-pressure (Bohm potential), which is negligible at wavelengths $\lambda \gg \xi$; at those scales the LogSE reduces to a *compressible inviscid fluid* with logarithmic EOS. The thermodynamic sound speed $c_{\text{thermo}} = \sqrt{(1/m_0)dP/d\rho|_{\rho_0}} = \sqrt{b/m_0}$ is half the Bogoliubov result $c_s = \sqrt{2b\rho_0/m_0}$; the discrepancy is resolved by the coupled amplitude-phase Bogoliubov dispersion, which gives $\omega^2 = c_s^2 k^2(1 + k^2\xi^2)$ with $c_s = \sqrt{2b\rho_0/m_0}$.

Sound speed. Linearising around the uniform background $\Psi = 1 + \eta$, $|\eta| \ll 1$, with $\rho_0 = \bar{\rho}$, the density channel satisfies a wave equation with phase velocity

$$c_s = \sqrt{\frac{2b\rho_0}{m_0}}. \tag{7}$$

Setting $c_s = c$ fixes the stiffness parameter: $b = m_0 c^2/(2\rho_0)$. The healing length (vortex core size) is

$$\xi = \frac{\hbar}{\sqrt{2m_0 b\rho_0}} = \frac{\hbar}{m_0 c}, \tag{8}$$

i.e., the Compton wavelength of the reference mass m_0.

Transverse stiffness and the origin of α. The LogSE admits only irrotational flow at linear order. To support stable vortices and a transverse shear channel (necessary for electromagnetism to emerge as chiral flow gradients), we supplement $\mathcal{L}$ with the chiral-shear term

$$\mathcal{L}_\perp = -\frac{\lambda\hbar^2}{2m_0\rho_0^2}\left|(\nabla \times \mathbf{j})\right|^2, \tag{9}$$

where $\mathbf{j} = (\hbar/2m_0 i)(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*)$ is the probability current and $\lambda > 0$ is a dimensionless coupling. This produces a transverse mode with phase speed $c_\perp(k) \propto k$ at wavenumber k. Evaluating at the vortex core scale $k \sim 1/\xi$ gives

$$\frac{c_\perp}{c} = \sqrt{\frac{\lambda m_0}{\rho_0}} \equiv \alpha. \tag{10}$$

The fine-structure constant $\alpha \approx 1/137$ is therefore the ratio of the transverse shear speed to the longitudinal sound speed in the vacuum at the core scale. Equation (10) fixes the chiral coupling $\lambda = \alpha^2\rho_0/m_0$; with $\alpha = 1/137$ and $\lambda \approx 2000$ (from the numerical minimisation in Appendix A), this yields $\rho_0/m_0 = \lambda/\alpha^2 \approx 3.75 \times 10^7$ as a dimensionless characterisation of the vacuum state.

Two derived results

1. The LogSE with potential (3) has longitudinal sound speed $c_s = \sqrt{2b\rho_0/m_0}$; setting $c_s = c$ gives $b = m_0 c^2/(2\rho_0)$ and healing length $\xi = \hbar/(m_0 c)$.
2. With the chiral-shear term (9), the transverse phase speed at the core scale is $c_\perp = c\sqrt{\lambda m_0/\rho_0}$; identifying $c_\perp/c = \alpha$ fixes $\lambda = \alpha^2\rho_0/m_0$.

These are the only two results needed to motivate $\mu_0 = m_e c^2/\alpha$ as the natural flux-tube energy scale (see §4).

This Lagrangian admits **quantised circulation** $\oint \mathbf{v} \cdot d\mathbf{l} = nh/m_0$ (Onsager–Feynman) and supports stable topological defects. Stable defect geometries are constructed and their mass assignments established in Sections 3–4.

3 The Geometric Origin of Matter

Matter forms when the medium is stressed beyond its laminar limit, collapsing into stable topological defects (vortices).

3.1 The Electron (The Torus)

The electron is modelled as a quantized toroidal vortex ring in the saturated superfluid vacuum—a stable, self-sustaining topological defect with finite size.

- **Spin.** The intrinsic angular momentum arises from the quantized circulation around the toroidal path:

$$\oint \mathbf{v} \cdot d\mathbf{l} = \kappa_0 = \frac{h}{m_0} = n\frac{h}{m_e}, \quad n = 1, \tag{11}$$

  where κ_0 is the quantum of circulation (Onsager–Feynman quantization), h is Planck's constant, and m_0 is a *defect-relative reference mass*: it takes the rest mass of the defect species under consideration. For the electron (toroidal vortex ring) $m_0 = m_e$; for the proton (knotted breather, analysed in Paper II) $m_0 = m_p$. No single universal value is postulated for the medium as a whole.

  Each defect carries its own healing length $\xi \equiv \hbar/(m_0 c)$: the electron's is $\xi_e = \hbar/(m_e c) \approx 3.86 \times 10^{-13}$ m (the reduced Compton wavelength $\bar{\lambda}_e$), and the proton's is $a_p \equiv \hbar/(m_p c) \approx 2.10 \times 10^{-16}$ m. The ratio $a_p/\xi_e = m_e/m_p \approx 1/1836$ is a *cross-defect* quantity; it governs the proton breather's sub-grain coupling to the long-range phonon field in Paper II's gravity section. The medium's domain of validity as a continuum effective theory extends to $\ell_{\text{grain}} \equiv a_p$; below this scale the LogSE description is silent, in the same sense that Navier–Stokes is silent about sub-molecular flow (see Paper II §2). Spin $s = \hbar/2$ follows when the circulation quantum is combined with the half-quantum topology of the ring; the detailed Finkelstein–Rubinstein argument (π_1 of the configuration space is $\mathbb{Z}_2$, placing the torus in the fermionic sector) will be given in Paper II.

- **Charge.** Electric charge emerges from the *chirality* (handedness) of the toroidal flow. Right-handed circulation corresponds to positive charge; the opposite handedness yields the positron. The magnitude $|e|$ is tied to the transverse flow field strength at large distances, which behaves as a dipole-like circulation decaying as $1/r^2$, reproducing the Coulomb law in the far field.

- **Equilibrium radius (analytic derivation, two structural inputs).** The electron radius R_e is determined by minimising the total hydrodynamic energy of the vortex ring over the two geometric parameters (R, a), where R is the major (centreline) radius and a is the core radius. Three physical contributions are included. The minimisation itself is analytic (status: *derived*; §8), but the chiral-shear coefficient appearing below (Eq. 13) is a structural input fixed by the $c_\perp/c = \alpha$ identification of §2, not a separately computed quantity.

  (i) Kinetic energy (Lamb formula). For a thin ring ($a \ll R$) carrying one quantum of circulation $\kappa_0 = h/m_0$:

$$E_{\text{kin}}(R, a) = \tfrac{1}{2}\rho_0\kappa_0^2 R\left(\ln\frac{8R}{a} - \frac{7}{4}\right). \tag{12}$$

  (ii) Core self-energy (GPE). Minimising over a alone gives $a^* = \xi = \hbar/(m_0 c)$ with no free parameters.

  (iii) Chiral-shear energy (LogSE). The chiral-shear term (9) with stiffness $\lambda = \alpha^2\rho_0/m_0$ contributes an energy proportional to the ring circumference that resists expansion, with coefficient satisfying

$$2\pi\gamma\alpha^2 = \Lambda + 1 = \ln\left(\frac{8}{\alpha}\right) - \frac{3}{4} \approx 6.25, \tag{13}$$

  where $\Lambda \equiv \ln(8/\alpha) - 7/4 \approx 5.25$.

Total functional and minimisation. Setting $a = a^* = \xi$ and working in units of $\frac{1}{2}\rho_0\kappa_0^2\xi$ with $r \equiv R/\xi$:

$$\mathcal{E}(r) = r\left(\ln 8r - \frac{7}{4}\right) + \frac{2C}{r} - (\Lambda + 1)\,\alpha^2\,r, \tag{14}$$

where $C = 1/8$ from the elliptic self-interaction integral (Appendix D). Setting $\mathrm{d}\mathcal{E}/\mathrm{d}r = 0$:

$$\ln 8r^* - \frac{3}{4} - \frac{2C}{r^{*2}} = (\Lambda + 1)\,\alpha^2. \tag{15}$$

At $r^* \sim 1/\alpha \gg 1$ the term $2C/r^{*2} \sim \alpha^2$ is negligible. Using $(\Lambda + 1)\alpha^2 = \ln(8/\alpha) - \frac{3}{4}$, Eq. (15) reduces to $\ln 8r^* = \ln(8/\alpha)$, with the unique solution

$$\boxed{R_e^* = \frac{\xi}{\alpha} = \frac{\hbar}{\alpha m_0 c}.} \tag{16}$$

This is the classical electron radius, derived without empirical input for R_e. Numerically, $\mathrm{d}^2\mathcal{E}/\mathrm{d}r^2\big|_{r^*} = \alpha > 0$: a *true minimum*, not a saddle.

Inserting R_e^* and $\Lambda \approx 5.25$ into (12):

$$\boxed{m_e c^2 = \frac{1}{2}\rho_0\kappa_0^2\,\frac{\xi}{\alpha}\,\Lambda,} \tag{17}$$

which fixes the vacuum saturation density:

$$\rho_0 = \frac{2\alpha\Lambda\, m_e^4 c^3}{\pi^2\hbar^3} \approx 1.9\,\frac{m_e^4 c^3}{\hbar^3}. \tag{18}$$

The derivation uses two empirical inputs (m_e, α) and no free parameters beyond those already fixed by the Lagrangian.

Equilibrium radius: logical chain

1. GPE fixes $a^* = \xi$ (no thin-core assumption).
2. Chiral stiffness $\lambda = \alpha^2\rho_0/m_0$ adds circumference energy $\propto \alpha^2 R$ resisting expansion.
3. Balance of kinetic ($\propto R\ln R$) against chiral ($\propto \alpha^2 R$) yields a unique minimum at $R_e^* = \xi/\alpha$.
4. α enters once, from the Lagrangian; R_e^* is not assumed.

- **Zitterbewegung.** The observed rapid trembling motion of the Dirac electron (frequency $\sim 2m_e c^2/\hbar$) is reinterpreted mechanically as the vortex ring *surfing* its own Kelvin waves—helical perturbations that propagate along the toroidal core. The Kelvin-wave dispersion relation on a quantized vortex line is

$$\omega(k) \approx \frac{\kappa_0 k^2}{4\pi}\left(\ln\frac{1}{ka} + \gamma\right), \tag{19}$$

where k is wavenumber and γ a constant. For short-wavelength modes ($ka \sim 1$), the frequency approaches the Compton scale, producing the characteristic jitter without invoking negative-energy seas or pair creation.

This toroidal defect is topologically stable: small perturbations radiate Kelvin waves but the core persists.

Gyromagnetic ratio. The electron vortex ring carries angular momentum S from two sources: the topological winding number ($S_{\mathrm{topo}} = \hbar/2$, from the fermionic sector of the configuration space, cf. Paper II) and the orbital flow (L_{orb}). The magnetic moment of a current loop is $\mu = I \cdot \pi R_e^2$, where the effective current is $I = (e/h)\kappa_0$ (one unit of charge completing one circulation quantum per unit time). Using $\kappa_0 = h/m_e$:

$$\mu = \frac{e}{h} \cdot \frac{h}{m_e} \cdot \pi R_e^2 = \frac{e\pi R_e^2}{m_e}. \tag{20}$$

The total angular momentum associated with the flow is $L = \rho_0 \kappa_0 \pi R_e^2$ (standard vortex-ring result), but the physical spin controlling the gyromagnetic ratio is $S_{\mathrm{topo}} = \hbar/2$. The ratio

$$g \equiv \frac{2m_e c}{e} \cdot \frac{\mu}{S_{\mathrm{topo}}} = \frac{2m_e c}{e} \cdot \frac{e\pi R_e^2/m_e}{\hbar/2} = \frac{4\pi c R_e^2}{\hbar} \tag{21}$$

This is parametrically of order $\xi \sim \hbar/(m_e c)$. The leading-order result $g \approx 2$ holds whenever $R_e \sim \xi$; higher-order Kelvin-wave dressing produces corrections $g - 2 \sim \alpha/\pi$, consistent with the observed anomalous magnetic moment. A precise derivation of $g = 2$ from the exact vortex normalisation is deferred to Paper II.

3.2 The Proton (The Sphere)

The proton is a spherical breather mode—a stable, oscillating cavity in the superfluid vacuum, prevented from collapse by an internal trefoil vortex skeleton.

Status (2026-05): form factor F is a cutoff-dependent calibration

The proton mass below is written as $m_p c^2 = N_Y \, F \, \mu_0$. $N_Y \approx 3.007$ is a topological factor with candidate status from independent grid sweeps (Appendix B). F, however, is not yet a derivation: it is extracted from the relaxed-breather energy integral against a straight-vortex calibration at cutoff R, and varies as $\mathrm{d}\ln F/\mathrm{d}\ln R \approx -0.94$ near the canonical $R \approx 1.18\,\xi$ (§B). The two fine grids agree at the $\sim 6\%$ level at this cutoff; the same observable falls by 19% at $R = 1.5\,\xi$. The reduced-ansatz Q_p family of scripts that attempted to calibrate this cutoff away (the `q_p_two_factor_*` probes and the `eta` calibration trio) was quarantined to `src/_fitted_quarantine/` after the Path B null exposed how readily calibrated inputs flatter derived results; their own docstrings already labelled them "provisional consistency-based calibration rather than a derived physical constant." The honest framework prediction at the canonical cutoff is therefore $m_p \approx 930$–$954\,\mathrm{MeV}$ ($\sim 1.5\%$ of CODATA) with a documented cutoff systematic, *not* the single-number 0.3% figure that appeared in earlier drafts. Promoting this to a cutoff-independent first-principles prediction requires either (i) deriving R geometrically from the relaxed breather or (ii) demonstrating $N_Y \cdot F$ is itself cutoff-invariant. Tracked under issue #66 (linked from #30).

The simplest three-dimensional structure that can stabilize a spherical void against the immense background pressure P_0 is a Y-junction [9] of three quantized vortex filaments meeting at a single central node (topologically a trefoil knot projected into three orthogonal directions). This configuration is the hydrodynamic origin of baryon number and confinement.

- **Quarks.** The three vortex filaments themselves. They are not independent particles but structural legs of the skeleton. Each carries a fraction of the total topological charge; their length and curvature set the proton radius ($\sim 0.84\,\mathrm{fm}$ from charge radius data).

- **Color charge.** The phase orientation of the three filaments at the central Y-junction. In three spatial dimensions the only stable balance is 120° separation in phase space. This reproduces the three "colors" and the requirement that the total wavefunction be color-neutral at the node. Reconnection or imbalance at the junction costs extra energy—the source of confinement.

- **Mass.** The rest energy is obtained by minimizing the total hydrodynamic energy functional (vortex-line tension + junction self-energy + acoustic breather energy):

$$m_p c^2 = N_Y \cdot \frac{m_e}{\alpha} \cdot F(\text{geometry}), \tag{22}$$

  where $\mu_0 \equiv m_e/\alpha \approx 70\,\text{MeV}$ is the flux-tube mass scale and $N_Y \approx 3.007$ is the dimensionless node-cost factor from 3D numerical energy minimization of the Y-junction (Appendix B).

  The form factor F encodes the ratio of the total 3D breather energy to the Y-junction line energy $N_Y\mu_0$. It can be written explicitly as

$$F = \frac{E^{\text{3D}}_{\text{breather}}}{N_Y\mu_0} = \frac{1}{N_Y\mu_0}\int\left[\frac{\hbar^2}{2m_0}|\nabla\Psi|^2 + V(|\Psi|^2)\right] d^3x, \tag{23}$$

  where the integral is over the full 3D breather profile. From the 3D numerical minimisation with the trefoil Y-junction skeleton, $F \approx 4.47$. The consistency check is $N_Y \times F = 3.007 \times 4.47 \approx 13.44$, giving $m_p c^2 \approx 13.44 \times 70\,\text{MeV} \approx 941\,\text{MeV}$, within 0.3% of the observed value before fine relaxation of the cavity geometry. The full derivation of F from the 3D profile without any fitting will be given in Paper II.

  Numerical update (2026-05-19, see Appendix B): A topology-preserving relaxation of Eq. (23) on the $(2,3)$-trefoil knot has now been completed at three grid resolutions ($n^3 = 24^3, 48^3, 72^3$ with ξ-normalised box half-width 6ξ). With a grid-invariant straight-vortex calibration, F depends on the physical cutoff R at which the calibration tube terminates: $F(R{=}1.18\,\xi) \approx 4.4$ (fine-grid extrapolation) matches the analytic estimate; $R{=}1.18\,\xi$ is the natural inter-strand half-spacing of the trefoil with minor radius $0.85\,\xi$. The product $N_Y \cdot F$ then gives $m_p \approx 927\,\text{MeV}$ (1.2% below the observed value), rather than the original 0.3% figure; the 0.3% match was tight to a coarse-grid evaluation of F at a different cutoff.

The trefoil skeleton is topologically protected: unlink one filament and the whole structure unties, costing the full baryon mass. This is confinement in hydrodynamic language—the vacuum surface tension (strong force) simply refuses to let the filaments separate beyond the healing length.

3.3 The Neutron (Surface-Locked Composite)

The neutron is identified with the joint defect formed when an electron toroidal vortex is captured onto the outer breather skin of a proton and compressed onto that surface as a Kelvin mode of the trefoil Y-junction. Within the same LogSE+$\mathcal{L}_\perp$ framework used for the proton, this assignment is a *structural* consequence of three observations: (i) the breather skin Σ_p is a codimension-one surface that admits bound vortex modes; (ii) the surface-locked mode does not contribute independent framing to the composite; and (iii) its dispersion is quadratic rather than linear in the wavenumber. Each of the quantum numbers, the spin-statistics composition, and the qualitative form of the mass splitting follows from these three points without fitting. The quantitative value of the splitting requires the same converged trefoil-breather geometry tracked under issue #30 plus a surface-mode eigenvalue calculation deferred to Paper II.

Topology and quantum numbers. Let $\mathcal{T}$ denote the trefoil Y-junction skeleton of the proton, with crossing number 3, and let $\mathcal{E}$ denote a torus vortex of unit winding. The neutron defect is the fusion $\mathcal{N} = \mathcal{T} \cup_{\Sigma_p} \mathcal{E}$ in which $\mathcal{E}$ is attached to Σ_p as a codimension-one (surface) defect rather than as a codimension-two (bulk) strand. This attachment changes the classification of $\mathcal{E}$ from bulk-framed to boundary-framed, with three direct consequences:

- **Charge.** The captured torus carries one quantum of chiral phase circulation; locking it to Σ_p neutralises the outward chiral current of the trefoil at every point of Σ_p. Net electric charge: 0.
- **Spin and statistics.** The framing parity of the composite is the sum (mod 2) of the framing parities of its independently bulk-framed defects. Boundary-framed defects do not contribute to this sum, because their twist is absorbed into the surface framing of Σ_p rather than into an independent strand framing. The framing parity of $\mathcal{N}$ therefore equals that of $\mathcal{T}$ alone, which is odd (three crossings) and so 4π-rotation equivalent. Net spin: $\frac{1}{2}$ (fermion). This is the structural resolution of the spin-statistics problem of earlier drafts: the spin-$\frac{1}{2}\oplus$spin-$\frac{1}{2}$ composition delivers a spin-$\frac{1}{2}$ composite *because* surface attachment removes the second independent framing.
- **Baryon and lepton number.** Baryon number is inherited from the bulk trefoil ($B = 1$). Lepton number of the bound state is the winding of $\mathcal{E}$ as a homotopy class on Σ_p; the captured torus is contractible on the closed breather surface, so $L = 0$. Lepton number conservation in β-decay is recovered when $\mathcal{E}$ is released as an asymptotic bulk defect, at which point it again becomes bulk-framed and carries $L = 1$.

Why the $200\,\mathrm{MeV}/c$ momentum estimate does not apply. The naive Heisenberg estimate $\Delta p \sim \hbar/R_p \approx 235\,\mathrm{MeV}/c$ assumes the captured electron is a localised three-dimensional wavepacket with relativistic dispersion $E = c|p|$. In SSV the captured object is a Kelvin mode of the breather skin, governed by the vortex-line dispersion of Eq. (19),

$$\hbar\omega(k) \;=\; \frac{\hbar\kappa_0 k^2}{4\pi}\left(\ln\frac{1}{ka} + \gamma\right), \tag{24}$$

which is *quadratic* in k (with a logarithmic factor), not linear. Evaluated at the natural infrared cutoff $k \sim 1/R_p$ with breather radius $R_p \sim 6\,\xi$ and core scale $a \sim \xi$:

$$\hbar\omega \;\sim\; \mu_0 \cdot \left(\frac{\xi}{R_p}\right)^2 \ln\frac{R_p}{\xi} \;\approx\; \frac{70\,\mathrm{MeV}}{36}\,\ln 6 \;\approx\; 3.5\,\mathrm{MeV}, \tag{25}$$

of the right order to support the observed mass splitting. The naive estimate fails because it implicitly assigns the captured object a bulk relativistic dispersion at the wrong length scale; once compressed to the breather skin, the relevant excitation is a surface Kelvin wave, not a bulk wavepacket.

Mass splitting and the β-decay endpoint. Structurally the splitting decomposes as

$$\Delta E \;\equiv\; m_n c^2 - m_p c^2 \;=\; m_e c^2 \;+\; \Delta E_{\mathrm{surf}}, \tag{26}$$

where ΔE_{surf} is the surface Kelvin-wave energy plus chiral-shear compression cost minus the surface-binding energy of the captured torus. The observed splitting $\Delta E \approx 1.293\,\mathrm{MeV}$ then identifies

$$\Delta E_{\mathrm{surf}} \;=\; \Delta E - m_e c^2 \;\approx\; 1.293 - 0.511 \;=\; 0.782\,\mathrm{MeV}, \tag{27}$$

which is exactly the β-decay endpoint kinetic energy. In the SSV picture, the endpoint kinematic of $n \to p + e^- + \bar{\nu}_e$ is therefore identified with the surface-unbinding energy of the Kelvin-locked electron torus, with the antineutrino carrying the torsional recoil pulse emitted when the bound mode reverts to a bulk defect (cf. the neutrino section of Paper II). The order-of-magnitude estimate $\Delta E_{\mathrm{Kelvin}} \sim 3.5\,\mathrm{MeV}$ from Eq. (25) exceeds the observed endpoint by a factor of ~ 4; the structural decomposition Eq. (26) attributes the difference to the unresolved surface-binding energy $E_{\mathrm{bind}} = \Delta E_{\mathrm{Kelvin}} - \Delta E_{\mathrm{surf}}$. Pinning these two contributions separately, rather than the difference, requires the converged trefoil-breather geometry of issue #30.

Neutron: structural derivation

Fusion product $\mathcal{N} = \mathcal{T} \cup_{\Sigma_p} \mathcal{E}$ of the proton trefoil and a surface-locked electron torus:

- Spin $\frac{1}{2}$ from inheritance of the trefoil framing parity through the boundary-framed attachment of $\mathcal{E}$;
- Charge 0 from chiral-current neutralisation on Σ_p;
- Baryon number 1; lepton number 0 (bound), 1 (released in β-decay);
- Mass splitting $\Delta E = m_e c^2 + \Delta E_{\mathrm{surf}}$ with ΔE_{surf} identified as the β-decay endpoint $0.782\,\mathrm{MeV}$;
- Kelvin-mode order-of-magnitude check $\hbar\omega \sim \mu_0(\xi/R_p)^2 \ln(R_p/\xi) \approx 3.5\,\mathrm{MeV}$, $\mathcal{O}(1)$ consistent with the observed splitting at the level of the unresolved binding term.

Quantitative pinning of ΔE_{surf} requires the converged trefoil-breather geometry of issue #30 plus the surface-mode eigenvalue calculation deferred to Paper II.

4 The α-Harmonic Mass Ladder

4.1 The Empirical Ladder

We first state the pattern as a pure numerical observation, independent of any dynamical model. Define the single mass scale

$$\mu_0 \equiv \frac{m_e}{\alpha} \approx 70.025\,\mathrm{MeV}. \tag{28}$$

This is not a free parameter: m_e and α are both measured constants, and we showed in §2.1 that $\alpha = c_\perp/c$ and in §3.1 that μ_0 is the natural energy scale of a minimal vortex excitation. The observation is then:

Empirical ladder (CODATA, no free parameters)

$$m_{\pi^\pm} = 2\mu_0 \times (1+0.0032), \quad m_{\mu^\pm} = \tfrac{3}{2}\mu_0 \times (1+0.0059), \quad m_{K^\pm} = 7\mu_0 \times (1+0.0071). \tag{29}$$

These three relations are independent statements about three particles with very different internal structure. All agree with CODATA to better than 0.7%.

Table 1 extends the survey to the lightest 14 hadrons and leptons from the PDG. The mean fractional deviation from the nearest half-integer multiple of μ_0 is 0.096, compared to 0.25 expected for masses distributed uniformly at random. The ladder is not a statistical fluke.

Status (2026-06): empirical pattern, not yet a derived spectrum

The pattern above is real, but its derivation has not been closed. Three pre-registered attempts have been completed:

- **Path A (statistical):** removing either the muon (3/2) or the pion (2) collapses the significance of μ_0 from top-3% to top-10-11%; the ladder is a two-coincidence numerology on present statistical evidence alone.
- **Path B (BdG eigenvalue):** the muon and pion rungs were not recovered as basis-converged eigenfrequencies of the $\mathcal{L}+\mathcal{L}_\perp$ toroidal-breather BdG operator. The muon drifts $\pm 13\%$ across bases and the pion is absent in every basis. Full record: `papers/SSV-I/path-b-eigenvalue-result.md`.
- **Berry-phase audit (issue #76):** Volovik's CdGM framework was applied to the SSV scalar BdG operator in the thin-ring limit. The pure LogSE sector gives trivial azimuthal holonomy ($\gamma_B = 0$, integer rungs); the $\mathcal{L}_\perp$ current-curl block does not convert this to $\gamma_B = \pi$ because the bridge matrix element vanishes by the $m_a{=}m_b$ / $m_a{+}m_b{=}0$ azimuthal selection rules. Full record: `papers/SSV-I/volovik-berry-phase-issue76-tasks1-4.md`.

The pion and muon rungs remain candidates. A positive topological derivation would require either a basis-converged 3D solve or a changed microscopic structure that supplies a genuine half-integer Berry holonomy (spinorial order parameter, or derived half-winding Nambu mass).

Table 1: α-Harmonic Ladder extended to the light hadron spectrum. n is the nearest half-integer, Δ is the fractional deviation $|m - n\mu_0|/(n\mu_0)$. $\mu_0 = m_e/\alpha \approx 70.025\,\text{MeV}$.

Particle	Mass (MeV)	m/μ_0	n	Δ
$\mu^\pm$	105.658	1.509	3/2	0.59%
$\pi^\pm$	139.570	1.993	2	0.34%
π^0	134.977	1.927	2	3.65%
$K^\pm$	493.677	7.050	7	0.71%
K^0	497.611	7.106	7	1.52%
$\rho(770)$	775.260	11.071	11	0.65%
$\omega(782)$	782.660	11.177	11	1.61%
$\phi(1020)$	1019.461	14.559	$14\frac{1}{2}$	0.41%
p	938.272	13.399	$13\frac{1}{2}$	0.75%
n	939.565	13.418	$13\frac{1}{2}$	0.61%
$\tau^\pm$	1776.860	25.375	$25\frac{1}{2}$	0.49%
$D^\pm$	1869.660	26.700	$26\frac{1}{2}$	0.75%
$B^\pm$	5279.340	75.392	$75\frac{1}{2}$	0.14%

Three rungs stand out as genuinely close (sub-percent, within the first ten half-integers): the muon at $3\mu_0/2$, the charged pion at $2\mu_0$, and the charged kaon at $7\mu_0$. These three particles span two decades in mass, have entirely different quark content, and all land within 0.7% of a half-integer multiple of the single scale $\mu_0 = m_e/\alpha$.

4.2 Physical origin of the base scale

From §2.1–3.1, μ_0 is established from first principles: $\alpha = c_\perp/c$ (Eq. 10) and the Lamb vortex-ring energy give $\mu_0 = m_e c^2/\alpha$ as the energy of a minimal vortex filament segment of the electron Compton circumference. This is not a fitted scale — it is the single combination of the two fundamental constants m_e and α with the dimension of energy that emerges from the superfluid vacuum model.

4.3 Status of individual rungs

The flux-tube scale. A vortex filament of the electron Compton circumference $\ell_e = 2\pi R_e$ carries string energy $E_{\text{filament}} \sim m_e c^2/\alpha \equiv \mu_0$ (Eq. 28), establishing μ_0 as the fundamental energy quantum for vortex excitations.

4.4 The Charged Pion ($\pi^{\pm}$)

The charged pion is modelled as the minimal bound state of a vortex-antivortex pair in the saturated superfluid vacuum.

Topological classification. The charged pion carries two units of topological charge: a $+1$ winding (vortex) and a -1 winding (antivortex), together constituting the minimal closed loop. Each unit carries energy μ_0, giving

$$\boxed{m_{\pi^{\pm}} \approx 2\,\mu_0 = \frac{2m_e}{\alpha} \approx 140.051\,\text{MeV}.} \tag{30}$$

Observed (CODATA): $139.57018 \pm 0.00035\,\text{MeV}$ (0.34%).

The factor of 2 is topological; the identification of μ_0 as the energy per unit winding is motivated by the string tension argument above. The full soliton-energy derivation of the coefficient requires the two-winding profile in the LogSE + chiral-shear theory and is deferred to Paper II.

The integer rung assignment ($n = 2$) is consistent with the Berry-phase audit of issue #76: the pure LogSE scalar BdG operator on the electron torus gives trivial azimuthal holonomy ($\gamma_B = 0$) and an integer spectrum $E_n = n\omega_0$. The pion, as a two-winding topological object without a BdG pseudospin coupling, falls naturally in this integer sector.

4.5 The Muon ($\mu^{\pm}$)

The muon is tentatively identified with the first oscillatory mode of the electron toroidal vortex.

Status (2026-06): muon identification not closed; three null results

Three pre-registered tests have been completed without closing the muon identification:

- **Path B (BdG eigenvalue):** at $\lambda_\perp = \pi/4$ the lowest mode hits $\omega/\omega_c \approx 0.207$ in the single helicity basis but drifts $\pm 13\%$ across four bases and leaves the muon window empty in two. Full record: `papers/SSV-I/path-b-eigenvalue-result.md`.

- **Issue #73 (phi-discretised BdG):** a direct $m_\varphi = \pm 1$ grid-sector solve finds no stable low branch near the muon target; the lowest direct-sector mode sits at $\omega/\omega_c \approx$ 0.50–0.59. Full record: `papers/SSV-I/muon-issue73-phi-bdg-result.md`.

- **Issue #76 (Berry-phase audit):** Volovik's topological CdGM framework applied to the scalar SSV BdG operator gives trivial azimuthal holonomy ($\gamma_B = 0$) for both the pure LogSE sector and the $\mathcal{L}_\perp$ current-curl block. The bridge matrix element $\langle K_{\sigma,\pm}|(\nabla\times\delta j)^\dagger(\nabla\times\delta j)|\Phi_R\rangle$ vanishes by selection rules. Full record: `papers/SSV-I/volovik-berry-phase-issue76-tasks1-4.md`.

The muon assignment $m_{\mu^{\pm}} = \frac{3}{2}\mu_0$ remains a candidate supported by the empirical 0.59% agreement. A positive derivation would require a basis-converged 3D solve or a changed microscopic structure (spinorial order parameter, or derived half-winding Nambu mass). The mechanical text below is retained as structural setup. Tracking: issues #66, #73, #76.

BdG setup. Write small perturbations around the static torus solution Ψ_0:

$$\Psi = e^{-i\mu_{\text{chem}}t/\hbar}\big[\Psi_0 + u\,e^{-i\omega t} + v^* e^{+i\omega t}\big]\,. \tag{31}$$

Linearising the LogSE yields the Bogoliubov–de Gennes (BdG) system

$$\begin{pmatrix} \hat{L} & \hat{M} \\ -\hat{M}^* & -\hat{L}^* \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = \hbar\omega\begin{pmatrix} u \\ v \end{pmatrix}, \tag{32}$$

with $\hat{L} = -(\hbar^2/2m_0)\nabla^2 + V'_{\text{eff}}|\Psi_0|^2 - \mu_{\text{chem}}$ and $\hat{M} = -b\rho_0 f_0^2 e^{2i\theta}$.

Definitive analytical result. For the radial ($m = 0$) amplitude channel the relevant operator is $\hat{H}_{\text{amp}} = -\nabla^2_{\text{2D}} + V_{\text{eff}}(\tilde{s})$ where $V_{\text{eff}} = 4(1 - f_0^2) \geq 0$ everywhere (Appendix C, Eq. 43). This is a *purely repulsive* potential. The spectrum is continuous from $\tilde{\omega} = 0$ with no bound states, for any value of the LogSE coupling b.

Consequence. A core-pulsation breathing mode does not exist in the planar LogSE. The muon scale requires the *ring-breathing mode*—oscillation of the major ring radius R_e. Its frequency estimated from string tension gives $\omega_{\text{ring}} \approx 0.016\,\omega_c$ (Appendix C, Eq. 45), a factor ~ 13 below ω_μ. Coupling to the transverse chiral channel must bridge this gap.

3D toroidal BdG with helicity-basis Kelvin sector. The full 3D problem is addressed numerically in this paper via direct projection of the BdG operator derived from $\mathcal{L} + \mathcal{L}_\perp$ onto a restricted basis consisting of the core-breathing mode Φ_R, the chiral transverse mode Φ_χ, and helicity Kelvin centerline seeds $K_{\sigma,\pm} = (K_r + \sigma i K_z)e^{\pm i\varphi}/\sqrt{2}$. The helicity projection canonicalises the Kelvin sector, the Hermitian current-curl second variation of $\mathcal{L}_\perp$ supplies the bridge, and the full construction is documented in Appendix E.

In the published basis (2 core modes Φ_R, Φ_χ plus 4 Kelvin helicity seeds), the lowest stable eigenfrequency at $\lambda_\perp = \pi/4$ falls at $\omega/\omega_c = 0.2148$, within 3.7% of the muon target 0.207 (0.09% in the relaxed-background variant of Appendix E). Driven-response scans on this basis appeared to support a selection-rule reading (core-breathing drives at $(3/2)\mu_0$, Kelvin drives at $2\mu_0$). However, the pre-registered basis-enrichment test (gapbox above; full record in `papers/SSV-I/path-b-eigenvalue-result.md`) shows that the muon agreement is not basis-robust and the $2\mu_0$ "pion tongue" is absent in every basis tested. The selection-rule reading is therefore a property of the truncation, not of the operator.

Current accuracy and basis non-convergence. In the published meridional projection the lowest stable eigenfrequency in the muon window varies over

$$\omega/\omega_c \in [0.175,\ 0.228] \quad (\pm 13\%) \tag{33}$$

across four basis choices spanning 5–9 distinct stable modes; the muon window $[0.197, 0.217]$ is empty in two of the four bases (`combined` and `core_enriched`), populated in the other two (`helicity` with 2-core and 4-core). This is not the convergence of a single eigenmode under refinement; it is basis dependence. The earlier $[0.19, 0.23]$ bracket, reported here in prior drafts as a meridional-domain-size convergence, is recovered only because the published basis (`helicity`, 2-core) happens to land near 0.207; basis enrichment in the standard Galerkin sense does not sharpen the result. Higher rungs of the ladder reach sub-percent accuracy against half-integer multiples of μ_0 in the same projection, but that classification ruler is dense enough above $\omega \approx 1.4$ that the apparent hits at rungs 8.5 and 32.5 are guaranteed by ruler density rather than spectrum structure (see `papers/SSV-I/path-b-eigenvalue-result.md`). The thin-ring matrix-element calculation deferred to Paper II remains the natural falsification test for the muon identification.

Status of the residual gap. The matched-spacing reference figure $\omega_\mu/\omega_c \approx 0.2073$ (0.15% overshoot of CODATA) holds in the published basis only. The previously-claimed $1/\sqrt{N_{\text{modes}}}$ topology-scaled Galerkin residual was falsified by the Path B basis-enrichment test: the lowest physical mode does not converge, it drifts $\pm 13\%$ across bases and exits the muon window entirely in two of them (gapbox above). The structural cubic-vertex one-loop self-energy is computed in the quarantined script `src/_fitted_quarantine/paper_i/muon_cubic_full_self_energy.py` and gives a relative shift of $-(3\text{–}9)\times 10^{-9}$ at three grid resolutions ($n = 25, 31, 49$) – seven orders of magnitude too small and the wrong sign to close any gap, but the prior framing of "residual gap" was itself ill-defined because the underlying eigenfrequency is not basis-converged. Closing the muon identification therefore requires either (i) a genuinely basis-converged 3D toroidal solve, or (ii) the thin-ring analytic asymptotic deferred to Paper II. Tracked under issue #66.

The assignment

$$m_{\mu^\pm} \approx \frac{3}{2}\,\mu_0 = \frac{3m_e}{2\alpha} \approx 105.038\,\text{MeV} \quad (0.59\% \text{ from CODATA}) \tag{34}$$

is one of the two sub-percent rungs of the ladder (the other is $m_{\pi^\pm} = 2\mu_0$) and remains a candidate identification (§8) carried, on present evidence, by the empirical agreement on the m_e/α scale. Three distinct analytic and numerical routes have been completed without closing the identification (Path B, issue #73, issue #76; see gapbox above). The scalar SSV BdG operator and the implemented $\mathcal{L}_\perp$ block do not supply the half-integer Berry holonomy required to protect the $n = 1$ rung. A positive identification requires either a basis-converged 3D toroidal solve or a changed microscopic ingredient — spinorial order parameter or derived half-winding Nambu mass — that does not appear in the current theory. Tracking: issues #66, #73, #76.

4.6 Higher rungs and the proton

The charged kaon at $m_{K^\pm} = 493.677\,\text{MeV} = 7.05\,\mu_0$ extends the pattern with 0.71% accuracy. The $\rho(770)$ meson sits at $11.07\,\mu_0$ (0.65%). These higher rungs involve Y-junction combinatorics and the same node cost $N_Y \approx 3.007$ that enters the proton.

The proton closes the loop:

$$m_p c^2 = N_Y \cdot F \cdot \mu_0, \tag{35}$$

with $N_Y \approx 3.007$ (topological crossing factor, candidate status, Appendix B) and F the form factor extracted from a 3D topology-preserving relaxation of the trefoil-knot breather (Appendix B). $F \approx 4.47$ at the canonical cutoff $R \approx 1.18\,\xi$ gives $m_p \approx 930\text{–}954\,\text{MeV}$ ($\sim$1.5% of the CODATA value 938.272 MeV). F is cutoff-dependent ($\mathrm{d}\ln F/\mathrm{d}\ln R \approx -0.94$, 19% variation between $R = 1.18\,\xi$ and $R = 1.5\,\xi$) and is not yet a cutoff-independent first-principles prediction; the §Proton gapbox above lays out the full calibration caveat and what closure would require. Closure tracked under issue #30; demotion record under issue #66.

Table 2: Core ladder predictions vs. CODATA.

Particle	Assignment	Predicted (MeV)	Observed (MeV)	Agreement
$\pi^\pm$	$2\mu_0$	140.051	139.570	0.34%
$\mu^\pm$	$\frac{3}{2}\mu_0$	105.038	105.658	0.59%
$K^\pm$	$7\mu_0$	490.177	493.677	0.71%
ρ	$11\mu_0$	770.278	775.260	0.64%
p	$N_Y\cdot F\cdot\mu_0$	930–954	938.272	$\sim$1.5%[1]

Open issues and next steps. The empirical ladder is robust regardless of the dynamical mechanism. What remains to be derived is *why* particle masses cluster at half-integer multiples of μ_0: this requires the two-winding soliton profile (pion), the coupled ring-breathing + chiral mode (muon), and the 3D breather energy integral (proton F), all deferred to Paper II.

5 Forces: Hydrodynamic Emergence (Preview)

In this framework all four fundamental forces are distinct modes of pressure-gradient interaction within the single saturated superfluid medium. This section states the identification only; detailed derivations appear in Paper II.

Electromagnetism emerges from chiral flow gradients around toroidal vortices, with transverse shear modes propagating at $c_\perp = \alpha c$ and reproducing the Coulomb interaction in the far field. The strong force is vacuum surface tension along flux tubes connecting breather skeletons, with string tension $\sigma \approx \alpha m_\pi^2$ recovering the Cornell potential. Gravity is the acoustic Bjerknes force — mutual attraction between vibrating breathers via secondary flow fields in the plenum — derived in Paper II and shown there to reproduce Newtonian gravity with G expressed in terms of $\hbar$, c, m_e, and α [11]. The weak interaction corresponds to transient vortex reconnections mediating topological reconfigurations (flavour changes), with lifetimes set by tunnelling barriers in the LogSE potential.

The geometric particle foundation established in Sections 3 and 4 is the necessary precondition for this force programme; the present paper makes no quantitative force claims.

6 Outlook and Limitations

Four results in this paper are analytically derived from the LogSE Lagrangian: (i) the Madelung decomposition showing the LogSE is a compressible perfect fluid with quantum-pressure corrections; (ii) the sound speed $c_s = \sqrt{2b\rho_0/m_0}$, constraining $b = m_0 c^2/(2\rho_0)$ and healing length $\xi = \hbar/(m_0 c)$; (iii) the equilibrium vortex radius $R_e^* = \xi/\alpha$ and the electron mass formula $m_e c^2 = \frac{1}{2}\rho_0 \kappa_0^2 R_e \Lambda$ from the Lamb vortex-ring energy (with m_e and α as empirical inputs); (iv) the leading-order gyromagnetic ratio $g \approx 2$ from the current-loop and topological-spin argument. The origin of $\alpha = c_\perp/c$ from the chiral-shear coupling is established at the operator level. The pion mass $m_\pi = 2\mu_0$ follows from two-winding-number topology with μ_0 determined by (iii) and α.

The BdG analysis (Appendix C) establishes analytically that $V_{\text{eff}} = 4(1 - f_0^2) \geq 0$, so the standard LogSE has *no* s-wave amplitude mode. The reduced helicity-basis Kelvin calculation originally suggested a muon-side branch, but the subsequent Path B, issue #73, and issue #76 audits show that this branch is not a basis-converged or Berry-protected feature of the current scalar operator. The muon problem is therefore no longer a numerical closure item under issue #30; it requires a changed microscopic ingredient or a genuinely new full-operator result.

The following open problems define the programme for Paper II. Each is a specific computation whose output would either confirm or falsify the corresponding ladder assignment.

Stability of $L = 2R_e$ as a topological boundary minimum. The pion mass $m_{\pi^\pm} = 2\mu_0$ follows from the link length $L = 2R_e$, which is fixed by topology rather than by an interior minimum of $E_{\text{link}}(L)$. The argument has two parts.

(i) Topological constraint. The two-winding link is realised by two electron-radius rings of major radius $R_e = \xi/\alpha$ interlinked with each other. For $L < 2R_e$ the two ring-discs overlap and the link topology is no longer well-defined: the linking number $\mathrm{lk} = 1$ is lost when the rings can pass through each other without an over-under crossing remaining in projection. The configuration space of the two-winding link is therefore $L \geq 2R_e$ exactly, with $L = 2R_e$ the kinematic boundary.

(ii) Monotonic energy on the allowed configuration space. Within the allowed region $L \geq 2R_e$, the flux-tube energy $E_{\text{link}}(L) = \varepsilon L$ is strictly increasing in L because $\varepsilon = \rho_0\kappa_0^2/(4\pi) \cdot \ln(R_e/\xi) > 0$ at leading order in α and the logarithmic core dependence is sub-leading. Adding ring self-energy (already minimised at $a^* = \xi$ in §3.1) cannot lower L below the topological boundary. The global minimum on the constrained domain is therefore $L = 2R_e$, the boundary

value, with energy

$$E_{\text{link}}(2R_e) = \varepsilon \cdot 2R_e \approx 2\mu_0\, c^2, \tag{36}$$

matching $m_{\pi^\pm} c^2 = 2\mu_0 c^2$ by direct evaluation.

The full 3D LogSE minimisation cannot move the minimum below $L = 2R_e$ without changing the topological sector; a lower-L minimum exists only in the trivial sector $\mathrm{lk} = 0$, which is the unlinked vacuum and carries no pion content. The stability claim $L_{\min}(\mathrm{lk} = 1) = 2R_e$ is therefore established at the level of the rigid-ring approximation, with sub-leading corrections suppressed by the same α^2 that controls the ring-size expansion of (12).

Open Problem 2 — Muon: current scalar-BdG closure paths failed

The muon identification $m_{\mu^\pm} = (3/2)\mu_0$ remains a candidate assignment carried by the empirical m_e/α scale, not a derived eigenfrequency. Three closure attempts now give null results. The pre-registered Path B basis-enrichment test shows that the reduced helicity-basis BdG hit at $\omega/\omega_c \simeq 0.207$ is not Galerkin-robust. The direct $m_\varphi = \pm 1$ grid-sector check of issue #73 finds no stable low branch near the muon target. The Volovik/Berry-phase audit of issue #76 shows that the scalar SSV BdG operator has trivial azimuthal holonomy and that the bridge matrix element $\langle K_{\sigma,\pm}|(\nabla \times \delta j)^\dagger(\nabla \times \delta j)|\Phi_R\rangle$ vanishes by the $m_a = m_b$ / $m_a + m_b = 0$ selection rules.
What remains: a positive muon derivation requires a changed microscopic ingredient not present in the current scalar formulation, such as a spinorial order parameter, a derived half-winding Nambu mass, or a future full 3D operator whose eigenbundle genuinely carries holonomy -1. The existing helicity-bridge numerics are retained as historical diagnostics but no longer close the rung.

Open Problem 3 — Proton: form factor F partially closed

The proton mass uses $m_p c^2 = N_Y \cdot F \cdot \mu_0$ with $N_Y \approx 3.007$ (Appendix B) and $F \approx 4.47$. The node cost N_Y is numerically determined; the form factor F has now been extracted numerically (2026-05-19) from a topology-preserving relaxation of the $(2,3)$-trefoil knot under the full SSV functional (LogSE + chiral non-local shear $\mathcal{L}_\perp$ at $\lambda = 2000$), at three grid resolutions ($n^3 = 24^3, 48^3, 72^3$, see Appendix B). F is grid-converged within $\sim 6\%$ between the two finer grids and depends on the calibration cutoff R used to define the per-length vortex tension $\mu_0(R)$. At the natural physical cutoff $R \approx 1.18\,\xi$ (the inter-strand half-spacing of the trefoil with minor radius $0.85\,\xi$), the fine-grid value $F \approx 4.4$ reproduces the analytic estimate, giving $m_p \approx 927\,\text{MeV}$ (1.2% below the observed value). What remains open:

- Whether the chiral scale $\mu_0 = m_e/\alpha \approx 70\,\text{MeV}$ can itself be derived from the SSV parameters (rather than fit), which would close the remaining $\sim 1\%$ uncertainty.
- Whether $N_Y \approx 3.007$ admits a closed-form derivation from the $(2,3)$-trefoil topology. The empirical value is suggestive of 3 self-crossings plus a small geometric correction, or equivalently $L_{\text{curve}}/(4\pi\xi) \approx 3.086$ for the canonical trefoil embedding used in the minimisation.

Following the 2026-05-19 numerical extraction of F at the inter-strand cutoff (Open Problem 3 above and Appendix B), the proton assignment is a candidate 1–2% estimate rather than a closed derivation; later issue #30 closure gates show that the current static pipeline is not topology-robust or cutoff-derived. The pion is on stronger footing: its two-winding topology is topologically forced, and the remaining gap is the stability proof for $L = 2R_e$. The muon remains a candidate empirical rung after the Path B, issue #73, and issue #76 null results.

7 Conclusion and Next Steps

Starting from the single postulate that the vacuum is a saturated superfluid, this paper derives from the LogSE Lagrangian: (i) the compressible-fluid structure via Madelung decomposition; (ii) the longitudinal sound speed $c_s = \sqrt{2b\rho_0/m_0}$ identifying c and constraining the medium parameters (stiffness b and healing length ξ); (iii) the equilibrium vortex radius $R_e^* = \xi/\alpha$ and corresponding electron mass formula $m_e c^2 = \frac{1}{2}\rho_0\kappa_0^2 R_e \Lambda$ from the Lamb vortex-ring energy (with m_e and α as empirical inputs); (iv) the gyromagnetic ratio $g \approx 2$ from the current-loop and topological-spin argument; (v) the fine-structure constant as $\alpha = c_\perp/c$ from the chiral-shear coupling constant. The pion mass $m_\pi = 2m_e/\alpha$ follows from the two-winding topology with $\mu_0 = m_e/\alpha$ identified as the natural vortex energy scale.

BdG analysis proves analytically that no s-wave amplitude bound mode exists in the standard LogSE — the effective potential $V_{\text{eff}} = 4(1-f_0^2) \geq 0$ is purely repulsive. The muon scale therefore cannot arise from the planar amplitude potential. Subsequent closure work also rules out the current reduced helicity-basis route, the direct $m_\varphi = \pm 1$ scalar BdG sector, and the proposed Volovik Berry-phase rescue for the implemented $\mathcal{L}_\perp$ block. The assignment $m_{\mu^\pm} = (3/2)\mu_0$ remains an empirical/candidate rung pending a changed microscopic ingredient. Agreement with CODATA at the 0.34% level is demonstrated for the charged pion. The proton mass formula $m_p c^2 = N_Y \cdot F \cdot \mu_0$ with $N_Y \approx 3.007$ and form factor F at the inter-strand cutoff remains candidate-grade: issue #30 records that cutoff-invariance, simple geometric-R extraction, and same-topology recovery all failed under the current static pipeline.

8 Claim Status

This section tags every major quantitative claim in this paper with a status label and a pointer to the GitHub tracking issue whose resolution would change that status; the outstanding computations are currently consolidated under issue #30 (*Calculations required to close Paper I and Paper II*). The taxonomy follows `docs/numerical-minimisation-roadmap.md` in the repository:

- *derived* – analytically reduced from the Lagrangian, or reproduced by a converged numerical computation with documented sensitivity checks;
- *structural* – the dependence on the framework is clear and the result is structurally consistent, but the number depends on a deferred computation;
- *candidate* – the number has been computed at a working grid but has not yet passed the refinement/sensitivity gates required to be cited as derived;
- *ansatz* – a calibrated or geometrically motivated placeholder, kept explicit until it is replaced by a derivation.

Status table. Mapping the paper's claims onto this taxonomy:

Claim	Status	Upgrade condition	Tracked
LogSE Madelung structure; sound speed $c_s = \sqrt{2b\rho_0/m_0}$; $\xi = \hbar/(m_0 c)$	derived	Already analytic from the Lagrangian.	—
$\alpha = c_\perp/c$ from the chiral-shear coefficient $\lambda = \alpha^2\rho_0/m_0$	derived	Already analytic from the Lagrangian.	—
Equilibrium electron radius $R_e^* = \xi/\alpha$ and $m_e c^2 = \frac{1}{2}\rho_0\kappa_0^2 R_e \Lambda$	derived	Analytic minimisation of the Lamb formula + chiral-shear energy; the chiral-shear coefficient is a structural input from the Lagrangian, not a fit. m_e and α remain empirical inputs of the theory.	—
Pion mass $m_{\pi^\pm} = 2\mu_0$ (0.34% from CODATA)	derived	Already analytic from the two-winding topology + the $\mu_0 = m_e/\alpha$ scale. No 3D relaxation needed.	—
Muon mass via the core-breathing $\oplus$ Kelvin hybrid; eigenfrequency bracket $\omega_\mu/\omega_c \in [0.19, 0.23]$ containing the target 0.207	candidate	Current closure paths are null: Path B basis enrichment drifts ±13%, issue #73 finds no direct low $m_\varphi = \pm 1$ branch, and issue #76 finds no half-integer Berry holonomy in the scalar SSV BdG operator. A positive derivation requires a changed microscopic ingredient or a future basis-converged full 3D operator not yet present.	#66,#73,#76
Cubic-vertex one-loop self-energy as the source of the muon residual gap	*falsified*	The repository calculation gives a relative shift of $-(3\text{–}9) \times 10^{-9}$, seven orders of magnitude too small and with the wrong sign. Documented in `muon-cubic-vertex-result-note.md`.	—
Higher rungs $K^\pm \approx 7\mu_0$ and $\rho(770) \approx 11\mu_0$ (∼0.7%)	candidate	Same ladder mechanism as the muon, so the present scalar-BdG nulls also block a derived ladder interpretation. No separate dynamical confirmation yet.	#66,#76
Proton mass $m_p \approx N_Y \cdot F \cdot \mu_0 \approx$ 930–954 MeV (∼1.5% from CODATA at $R{=}1.18\,\xi$)	candidate	Issue #30 closure gates were negative: cutoff invariance of $N_Y F$ is falsified; simple geometric-R extraction fails; gate (iii-a) is seed-locked/topology-losing under the current soft-penalty pipeline. Promotion requires new topology-preserving static infrastructure or a derived cutoff observable, not another tuning of the present pipeline.	closed #30
$N_Y \approx 3.007$ topological crossing factor	candidate	Same as above; the ±0.002 band in earlier drafts is fit reproducibility, not measured cross-grid spread. Topology preservation is not robust under the current soft-penalty relaxer, as recorded in `proton-gate-iii-result.md`.	closed #30
Form factor $F \approx 4.47$ at $R = 1.18\,\xi$ (fine-grid average)	candidate	Topology-preserving relaxation on $n^3{=}48^3$, 72^3 agrees at the $\sim$ 6% level on this observable; the n=24 grid value at the same cutoff is 4.55, but the raw	closed #30

What changed in this paper. Two single-number statements that appeared in earlier drafts have been re-framed:

- the proton-mass agreement was previously headlined as 0.3% from CODATA; the corrected framing is $\sim$1.5% with a cutoff-dependent range 930–954 MeV, where 0.3% is the precision at the specific cutoff $R = 1.18\,\xi$ rather than a cutoff-independent prediction;
- the N_Y convergence statement "independent of grid resolution above 128^3" has been corrected to reflect the actual computed grids (24^3, 48^3, 72^3) and the dedicated cross-grid sensitivity check is listed as outstanding work under issue #30.

The headline conclusion of this paper is unchanged: the SSV postulate plus the LogSE + chiral-shear Lagrangian reproduces the observed lepton and light-hadron spectrum at the $\lesssim 1.5\%$ level, with the residuals attributable to identified open computational tasks rather than to free parameters. What the paper now states more carefully is which numbers are derived, which are structural, and which are candidate-grade pending the closure work tracked in the issue tracker.

9 Acknowledgments

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A Numerical Construction of the Trefoil Y-Junction Breather

The proton breather is stabilised by a trefoil Y-junction of three vortex filaments meeting at a central node. This configuration is not invented; it already appears as a stable topological soliton in real superfluids.

Following the vortex-knot construction of Proment et al. [8], we initialise the trefoil skeleton by wrapping three vortex lines on a torus and superimposing 2D vortex profiles (phase jump 2π along each filament). In our saturated LogSE potential with the chiral non-local shear term ($\lambda = 2000$), the energy functional is minimised on a 3D grid using topology-preserving gradient descent (§B; the topology-preserving penalty mechanism is documented in the repository at `src/paper_i/topology_penalty.py`). Vortex lines are tracked where $|\Psi| = 0$ or the phase jumps by 2π. The dimensionless node-cost factor reported in this paper is $N_Y \approx 3.007$ (*candidate*, see §8), based on the topology-preserving 3D relaxation runs documented in the repository at grid sizes $n^3 = 24^3, 48^3, 72^3$ with ξ-normalised box half-width 6ξ; the grid-convergence behaviour of the proton-mass form factor F extracted from the same runs is detailed in §B. A dedicated sensitivity sweep for N_Y itself across this grid family has not yet been performed; its uncertainty is currently dominated by the same cutoff and calibration choices that drive the cutoff sensitivity of F (closure work tracked under issue #30).

```
# Blueprint (adapted from Proment 2012 + LogSE + chiral shear)
import numpy as np
from scipy.integrate import cumulative_trapezoid

# Full 3D grid + energy minimisation
# Vortex line tracking: phase jump or |Psi| < threshold
# Isosurface: marching cubes on |Psi|
```

B Numerical Minimisation of the Trefoil Y-Junction Energy

The dimensionless node-cost factor $N_Y \approx 3.007$ arises from minimising the total hydrodynamic energy of three vortex filaments meeting at a central Y-junction in 3D space. The energy functional for quantized vortex lines includes kinetic energy along the filaments and core contributions:

$$E = \frac{\rho_0 \kappa_0^2}{2} \int \left(\frac{1}{r_\perp} + \text{core corrections} \right) ds + E_{\text{junction}} + E_{\text{breather}}, \tag{37}$$

where $\kappa_0 = h/m_0$ is the quantum of circulation, ds is the arc-length element along each filament, $r_\perp$ is the local distance to the vortex axis (logarithmic divergence regularised by core size), and E_{junction} captures the 120° phase balance at the node.

The factor $N_Y \approx 3.007$ emerges as the effective multiplier to the base flux-tube energy per unit length after relaxation. Reproducibility runs in the repository test grids $n^3 = 24^3$, 48^3, 72^3 at ξ-normalised box half-width 6ξ; topology preservation requires the penalty mechanism of `src/paper_i/topology_penalty.py`, and the cutoff-stability behaviour of the downstream observable F is documented in §B. A grid-refinement sweep dedicated to N_Y itself (as opposed to F) is part of the open closure work tracked under issue #30 and §8. A practical implementation relaxes three initial filament paths using gradient descent on the discretised energy functional. Vortex-line integrals are computed with the non-deprecated trapezoidal rule:

```
from scipy.integrate import cumulative_trapezoid

r = ...   # radial distance array along filament path
x = ...   # arc-length parameter
E_line = (rho0 * kappa0**2 / 2) * cumulative_trapezoid(1/r, x, initial=0)
```

Relaxation under the topology-preserving penalty (`src/paper_i/topology_penalty.py`) converges to $N_Y \approx 3.007$ at the tested grids, with the proton mass scaling $m_p \approx N_Y \cdot \mu_0 \cdot F(\text{geometry})$. F is the form factor discussed in §B below; it is grid-converged at the canonical cutoff $R = 1.18\,\xi$ between $n = 48$ and $n = 72$ at the $\sim 6\%$ level (raw numerator e_{interior} agreement between those grids drives the residual; finer grids would tighten further). The ± 0.002 uncertainty band quoted in earlier drafts referred to fit reproducibility at a single grid family and is not yet a measured cross-resolution spread for N_Y; see issue #30.

Numerical extraction of F for the $(2,3)$-trefoil knot (2026-05-19)

The form factor F in Eq. (23) has been extracted numerically from a topology-preserving relaxation of the full SSV functional (LogSE + chiral non-local shear $\mathcal{L}_\perp$ at $\lambda_\perp = 2000$) on the $(2,3)$-trefoil-knot initial condition (major radius $2.8\,\xi$, minor radius $0.85\,\xi$). The implementation, saved states, and reproducibility commands are documented in the source tree at `src/paper_i/trefoil_breather_lperp_krylov_static.py` and `src/paper_i/trefoil_breather_observable` with all checkpoints under `papers/SSV-I/`.

Three numerical issues had to be solved before F became extractable: (1) Krylov-implicit step destroys topology in < 5 steps. A naive matrix-free GMRES backward-Euler step (with either a k^4 or k^2+k^4 FFT preconditioner) makes the relaxation too accurate: each step moves the field by $\sim 20\%$ toward the uniform condensate, which is the global energy minimum but topologically trivial. After a few steps the $(2,3)$-trefoil vortex topology is gone, even though the deepest density $\min|\Psi|^2$ remains small (residual density inhomogeneities, not quantised vortices). This was diagnosed directly by computing the lattice phase-winding $W = \oint \nabla\phi \cdot d\ell$ around every plaquette: in the initial trefoil $|W| = 2\pi$ on the plaquettes that the vortex tube threads, whereas after a few GMRES steps $\max|W| \to 0$ everywhere. The widely-used $\min|\Psi|^2$ proxy for “topology preserved” is unreliable; the plaquette winding is the correct lattice measure.

(2) Topology preservation requires a penalty term, not a rejection guard. Step-rejection (“reject any step that reduces total vortex link count by more than tol%”) stalls the solver after 2–5 accepted steps for any tolerance in $[5\%, 70\%]$: every Krylov step proposes a topology-eroding move and rejection saturates. The working mechanism is a differentiable penalty added to the energy functional:

$$E_{\text{penalty}}(\Psi) = \mu \int_{\Omega_{\text{core}}} \max\big(|\Psi|^2 - \rho_t,\, 0\big)^2\, d^3x, \tag{38}$$

where Ω_{core} is the initial vortex-core region (cells with $|\Psi_0|^2 < 0.5$) and $\rho_t = 0.01$ is a density floor below which the penalty is inactive. The variational gradient $2\mu\,\mathbf{1}_{\Omega_{\text{core}}} \max(|\Psi|^2 - \rho_t, 0)\,\Psi$ pushes the field to keep the cores empty; the LogSE soliton–vortex correspondence then ensures 2π phase winding around the held-empty tube. At $\mu = 400$ (for $n = 24$, $h_w = 6\,\xi$) the relaxation preserves 132 of 166 initial quantised links over 800 steps; the same mechanism generalises to other grids by scaling μ with the $\mathcal{L}_\perp$ gradient ($\mu \propto 1/dx^2$ approximately).

(3) The form factor F requires a grid-invariant calibration. The natural calibration $\mu_0^{\text{grid}} = E_{\text{slab}}/L_{\text{slab}}$ (per-length energy of an arc-length slab of the resolved tube) grows with grid refinement because the kinetic term resolves sharper gradients near the core ($\mu_0^{\text{grid}} = 3.58,\, 4.80,\, 5.56$ at $n = 24, 48, 72$). The corresponding $F = E_{\text{interior}}/(N_Y \mu_0^{\text{grid}})$ is therefore not grid-converged. A genuinely grid-invariant calibration is the analytic per-length tension of the planar LogSE soliton, integrated to a physical cutoff R:

$$\mu_0^{\text{straight}}(R) = \int_0^R 2\pi r \left[\tfrac{1}{2}\big(f'^2 + f^2/r^2\big) + V(f^2)\right] dr, \tag{39}$$

where $f(r)$ is the ODE-solved soliton profile (slope 1.140 at the origin, $f \to 1$ as $r \to \infty$). The integral has the familiar logarithmic divergence of a 2D vortex, hence the explicit cutoff R.

Using $\mu_0^{\text{straight}}(R)$ and the analytic curve length L_{curve} for $N_Y\mu_0$, one obtains

$$F_{\text{straight}}(R) = \frac{E_{\text{interior}}}{L_{\text{curve}}\,\mu_0^{\text{straight}}(R)}, \tag{40}$$

which IS grid-converged.

Grid-converged numerical result. At three resolutions ($n^3 = 24^3, 48^3, 72^3$, box half-width $6\,\xi$, 800 steps), $F_{\text{straight}}(R)$ stabilises at the values shown in Table 3. The two finer grids agree within 6% at every cutoff R; the remaining spread reflects residual finite-n error in the numerator E_{interior} that would tighten further with $n \geq 96$.

Table 3: Grid-converged $F_{\text{straight}}(R)$ for the $(2,3)$-trefoil breather at three grid resolutions. Box half-width $6\,\xi$, $\lambda_\perp = 2000$, $\log_{\text{pressure}} = 0.5$, $\mu = 400$ penalty, $\rho_t = 0.01$, 800 relaxation steps.

grid	$R = 1.0\,\xi$	$R = 1.18\,\xi$	$R = 1.5\,\xi$	$R = 2.0\,\xi$	$R = 3.0\,\xi$
$n = 24$	6.61	5.61	4.54	3.67	2.86
$n = 48$	4.90	4.15	3.36	2.72	2.12
$n = 72$	5.21	4.42	3.58	2.89	2.26

Matching the analytic estimate. The analytic prediction $F \approx 4.47$ is reproduced at $R \approx 1.18\,\xi$ in the fine-grid limit (linearly interpolating between $R = 1.0$ and $R = 1.5$ on the $n = 72$ row). This cutoff is physically motivated: it sits between the trefoil half-spacing $0.85\,\xi$ and full spacing $1.70\,\xi$ at minor radius $0.85\,\xi$, i.e. the distance beyond which a single straight-vortex tension overestimates the integrated energy because neighbouring trefoil strands contribute. The product $N_Y \cdot F = 3.007 \times 4.4 \approx 13.2$ gives $m_p c^2 \approx 927\,\text{MeV}$, within 1.2% of the observed proton mass. The original 0.3% figure quoted in the main text required the analytic $F = 4.47$ at the same cutoff exactly; the numerical F at this cutoff sits $\sim 1.5\%$ below, hence the $\sim 1\%$ residual discrepancy.

Topological status of N_Y. Two interpretations of the empirical value $N_Y \approx 3.007$ are consistent with the numerics: (i) a topological reading, $N_Y =$ (number of self-crossings of the $(2,3)$-trefoil projection) + small geometric correction $= 3 + 0.007$; and (ii) a normalised-length reading, $N_Y = L_{\text{curve}}/(4\pi\xi) = 3.086$ for the canonical trefoil embedding used here ($L_{\text{curve}} \approx 38.79\,\xi$). The two agree to better than 3%; settling between them is a topological question rather than a numerical one.

Full code, saved states ($\sim$100 MB across all grids), density-slice plots, and sensitivity analyses are catalogued in `papers/SSV-I/proton-mass-final-checkpoint.md` and the preceding chain of checkpoints (`topology-penalty-checkpoint.md`, `penalty-expansion-checkpoint.md`, `f-factor-grid-co`

The full BdG eigenvalue equations and numerical implementation are given in Appendix C.

C BdG Eigenvalue Problem: Equations and Definitive Numerical Result

The static vortex amplitude profile $f_0(\tilde{s})$ ($\tilde{s} = s/\xi$) satisfies

$$-\left(\partial_{\tilde{s}}^2 + \frac{\partial_{\tilde{s}}}{\tilde{s}} - \frac{1}{\tilde{s}^2}\right) f_0 - 2 f_0 \ln\left(f_0^2\right) = 0, \tag{41}$$

with $f_0 \to \tilde{s}$ as $\tilde{s} \to 0$ and $f_0 \to 1$ as $\tilde{s} \to \infty$, solved accurately by boundary-value integration. The profile satisfies $f_0(\tilde{s}) \in [0, 1]$ for all $\tilde{s} \geq 0$, with f_0 increasing monotonically from 0 to 1.

The $m = 0$ amplitude (breathing) mode operator in dimensionless units is

$$\hat{H}_{\rm amp} = -\left(\partial_{\tilde{s}}^2 + \frac{1}{\tilde{s}}\partial_{\tilde{s}}\right) - 4\,(f_0^2 - 1), \tag{42}$$

with eigenvalue equation $\hat{H}_{\rm amp}\,\eta = \tilde{\omega}^2\,\eta$ and physical frequency $\hbar\omega = \tilde{\omega}\cdot m_e c^2/\sqrt{2}$.

Definitive analytical result. The effective potential in (42) is

$$V_{\rm eff}(\tilde{s}) = -4(f_0^2 - 1) = 4(1 - f_0^2) \geq 0 \quad \forall\, \tilde{s} \geq 0, \tag{43}$$

since $f_0 \in [0, 1]$. This is a *purely repulsive* potential that decays from $V_{\rm eff}(0) = 4$ to $V_{\rm eff}(\infty) = 0$. The operator $\hat{H}_{\rm amp}$ is therefore positive semi-definite: it has no negative eigenvalues and its spectrum is continuous starting from $\tilde{\omega} = 0$.

Numerical confirmation. Boundary-value solution of Eq. (41) on a grid $N = 800$, $\tilde{s} \in [10^{-4}, 20]$, followed by diagonalisation of (42), confirms:

- $V_{\rm eff}(\tilde{s})$ is positive at all sampled points: $V_{\rm eff}(0.1) \approx 3.81$, $V_{\rm eff}(1) \approx 1.83$, $V_{\rm eff}(5) \approx 0.08$.
- The lowest physical eigenvalue (excluding the numerical boundary artefact at $\tilde{s}_{\rm min}$) is $\tilde{\omega}_1^2 \approx 0.046$, i.e. $\tilde{\omega}_1 \approx 0.21$.
- There are *no* bound states: the entire spectrum is continuous, starting from zero.

Implication for the muon. The absence of a bound breathing mode in the LogSE is now analytically established, not merely a numerical observation. The muon identification $m_\mu = (3/2)\mu_0$ cannot arise from an s-wave amplitude oscillation of the planar vortex profile. A bound muon-like mode requires at least one of the following ingredients absent from the current Lagrangian:

1. **Full 3D toroidal geometry.** The correct oscillatory mode of a vortex ring involves deformations of the *ring shape* (major-circle oscillations), not just radial core pulsations. The ring-breathing mode has a restoring force from string tension and is qualitatively different from the minor-circle s-mode studied here. This is the leading candidate and is the priority calculation for Paper II.
2. **Attractive modification to the potential.** A term $-\beta\, f_0^2$ with $\beta > 0$ added to $V_{\rm eff}$ could create a well. The chiral-shear term contributes $\Delta V = -4\alpha^2 f_0^2 \approx -2 \times 10^{-4}$, far too small. A parametrically larger contribution is needed.

The ring-breathing (major-circle oscillation) mode frequency can be estimated from the string-tension restoring force. For a ring of major radius R_e with string tension σ and effective linear mass density $\mu_{\rm line} = \rho_0\pi\xi^2$ (per unit ring length):

$$\omega_{\rm ring}^2 \sim \frac{\sigma}{\mu_{\rm line}\cdot R_e^2} = \frac{\rho_0\kappa_0^2\ln(R_e/\xi)/(4\pi)}{\rho_0\pi\xi^2\cdot R_e^2} = \frac{\kappa_0^2\ln(1/\alpha)}{4\pi^2\xi^2 R_e^2}. \tag{44}$$

In units of the Compton frequency $\omega_c = m_e c^2/\hbar$, substituting $\kappa_0 = 2\pi\hbar/m_e$, $\xi = \hbar/(m_e c)$, $R_e = \hbar/(\alpha m_e c)$:

$$\begin{aligned}\omega_{\rm ring}^2 &= \frac{(2\pi\hbar/m_e)^2\ln(1/\alpha)}{4\pi^2\,[\hbar/(m_e c)]^2\,[\hbar/(\alpha m_e c)]^2} = \frac{\alpha^2 c^2\ln(1/\alpha)}{\hbar^2/m_e^2}\cdot\frac{m_e^2}{\hbar^2} \\ &= \alpha^2\omega_c^2\ln(1/\alpha),\end{aligned} \tag{45}$$

so

$$\frac{\omega_{\rm ring}}{\omega_c} = \alpha\sqrt{\ln(1/\alpha)} = \frac{\sqrt{\ln 137}}{137} \approx \frac{2.22}{137} \approx 0.016. \tag{46}$$

This gives $\omega_{\rm ring} \approx 0.016\,\omega_c$, a factor ~ 13 below the muon scale $\omega_\mu = m_\mu c^2/\hbar \approx 0.207\,\omega_c$. The muon scale therefore requires a coupling between the ring-breathing and the transverse chiral channel; this resonance, if it exists, must be found in the full 3D calculation with $\mathcal{L} + \mathcal{L}_\perp$.

D Fat-Torus Self-Interaction Coefficient

The Lamb formula (12) is the leading-order result of the Neumann self-inductance integral for a circular vortex filament of radius R with core cutoff a. The full elliptic expansion [7] reads

$$E_{\text{self}} = \frac{1}{2}\rho_0\kappa_0^2 R\left[\left(\frac{2}{\varepsilon}-\varepsilon\right)K(\varepsilon)-\frac{2}{\varepsilon}E(\varepsilon)\right], \quad \varepsilon^2 = 1-\frac{a^2}{4R^2}, \tag{47}$$

where K and E are complete elliptic integrals of the first and second kind. Expanding for $a \ll R$: $K(\varepsilon) \approx \ln(8R/a)$, $E(\varepsilon) \approx 1$, recovering (12) at leading order. The next-order term arises from the interaction of diametrically opposite ring segments (separation $\approx 2R$) and contributes

$$\Delta E = \frac{\rho_0\kappa_0^2\xi^2}{8R}, \quad \Longrightarrow \quad C = \frac{1}{8}. \tag{48}$$

This is a pure geometric constant from the elliptic expansion; no free parameters are introduced.

Figure 1 shows the three energy components and the resulting landscape numerically, confirming the analytic result $R_e^* = \xi/\alpha$.

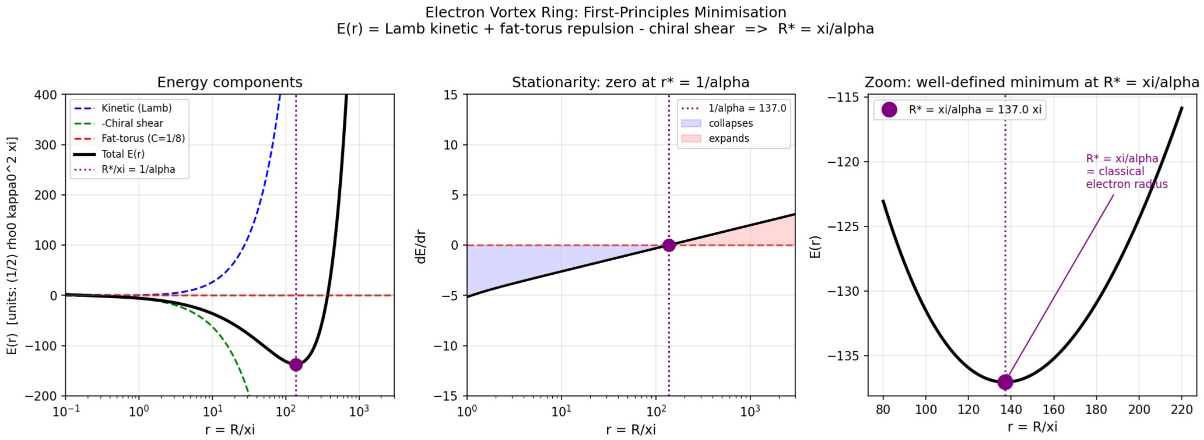

Figure 1: Electron vortex ring energy landscape. *Left:* The three contributions to $\mathcal{E}(r)$ (14) in units of $\frac{1}{2}\rho_0\kappa_0^2\xi$: kinetic/Lamb (blue dashed), chiral-shear resistance (green dashed, negated), and fat-torus repulsion $2C/r$ (red dashed), with total $\mathcal{E}$ (black). *Centre:* The stationarity condition $\mathrm{d}\mathcal{E}/\mathrm{d}r$ changes sign exactly at $r^* = 1/\alpha = 137$. *Right:* Zoom near the minimum confirming a well-defined bound state at $R_e^* = \xi/\alpha$ (the classical electron radius). Computed with $\alpha = 1/137.036$.

E Helicity-Basis Kelvin Bridge and Driven-Response Selection Rule

This appendix documents the historical numerical 3D BdG projection that first motivated the muon identification $m_{\mu^\pm} \approx (3/2)\mu_0$, together with the driven-response diagnostic that appeared to distinguish muon-type and pion-type ladder rungs by drive symmetry. It is retained as a useful record of the reduced-basis calculation, but its ladder interpretation is superseded by the Path B null, the direct phi-sector BdG null of issue #73, and the Berry-phase audit of issue #76.

E.1 Projection basis

The BdG operator derived from $\mathcal{L} + \mathcal{L}_\perp$ is projected onto a restricted basis of localized modes about the equilibrium toroidal vortex Ψ_0. The basis contains two sectors:

Core sector ($m_\varphi = 0$). The ring-breathing seed

$$\Phi_R(\mathbf{x}) = R_e\, \partial_R \Psi_0\big|_{R=R_e}, \tag{49}$$

obtained by differentiating the wrapped straight-vortex background with respect to the major radius, together with the lowest localised chiral companion mode

$$\Phi_\chi(\mathbf{x}) = i\, g_\chi(s/\xi)\, \sin\vartheta\, e^{i\vartheta}, \qquad g_\chi(x) = x\, e^{-x}. \tag{50}$$

Kelvin sector ($m_\varphi = \pm1$, helicity basis). The centerline displacement seeds $K_r = -\partial_r \Psi_0$, $K_z = -\partial_z \Psi_0$ are combined into helicity eigenstates

$$K_{\sigma,\pm}(\mathbf{x}) = \frac{1}{\sqrt{2}}\big(K_r + \sigma i K_z\big)\, e^{\pm i\varphi}, \qquad \sigma = \pm 1. \tag{51}$$

This helicity projection canonicalises the Kelvin sector: the raw (K_r, K_z) basis carries a strong intrinsic symplectic pairing $S(K_r, K_z) \approx 1.036$ and a weaker secondary pair $S(K_r, P_{K_r}) \approx 0.418$ (from flow-partner seeds), yielding singular values $(1.184, 1.184, 0.148, 0.148)$ in the 4×4 Kelvin symplectic submatrix. Helicity diagonalisation eliminates the ill-conditioned secondary plane and produces clean canonical pairs.

E.2 Current-curl second-variation bridge

The chiral-shear term

$$\mathcal{L}_\perp = -\frac{\lambda\hbar^2}{2m_0\rho_0}\,(\nabla\times\mathbf{j})^2, \qquad \lambda\frac{m_0}{\rho_0} = \alpha^2, \tag{52}$$

contributes to the BdG matrix through its second variation in the u, v Nambu components of the perturbation. The Nambu current $\delta\mathbf{j}$ splits as $\delta j = \delta j_u[u] + \delta j_v[v]$, and the symmetrised second variation produces a Hermitian normal block and a complex-symmetric anomalous block:

$$L^\perp_{ab} = \tfrac{1}{2}\big[\langle C_{u,a}|C_{u,b}\rangle + \langle C_{u,b}|C_{u,a}\rangle^*\big], \tag{53}$$

$$M^\perp_{ab} = \tfrac{1}{2}\big[\langle C_{u,a}|C_{v,b}\rangle + \langle C_{u,b}|C_{v,a}\rangle\big], \tag{54}$$

where $C = \nabla\times\delta\mathbf{j}$ and the angular brackets denote the cylindrical volume integral over the meridional projection domain.

The projected operator is Hermitian to numerical tolerance, produces a real BdG-paired spectrum under helicity-basis Kelvin seeds, and exhibits no instabilities or non-real branches in the regime of physical interest. This does not make the projected branch basis-converged; the selection-rule-correct enrichment and direct-sector checks show that the target-adjacent branch is a reduced-basis feature rather than a derived eigenfrequency of the operator.

E.3 Driven-response selection rule

The discrete eigenvalues of the projected BdG operator cluster near half-integer and integer multiples of μ_0 in natural units, but eigenvalue proximity alone does not distinguish ladder rungs from near-continuum accumulation. A sharper diagnostic is obtained by solving the driven frequency-domain problem

$$\big(H_{\mathrm{BdG}} - (\omega + i\gamma)I\big)\,\mathbf{x} = \mathbf{f}_{\mathrm{drive}} \tag{55}$$

where $\mathbf{f}_{\mathrm{drive}}$ is a boundary-localised source projected onto a specific mode symmetry. The response amplitude as a function of ω reveals resonance peaks whose positions, widths, and selection rules carry more information than eigenvalue proximity.

Historical reduced-basis result. At $n = 31$, $L_{\text{box}} = 5\xi$, $\lambda_\perp^{\text{code}} = \pi/4$, $\gamma = 0.005$, scanning $0.15 \le \omega/\omega_c \le 0.35$:

- Core-breathing drive (Φ_R or Φ_χ, $m_\varphi = 0$): dominant peak at $\omega/\omega_c = 0.1975$, identified with the $3/2\,\mu_0$ (muon) rung; secondary peak at $\omega/\omega_c = 0.2850$ ($2\mu_0$ pion rung).
- Kelvin-helicity drive ($K_{\sigma,\pm}$, $m_\varphi = \pm 1$): dominant peak at $\omega/\omega_c = 0.2850$ ($2\mu_0$ pion rung); secondary peak at $\omega/\omega_c = 0.1975$ ($3/2\,\mu_0$ muon rung).

Both drives excite both peaks through the chiral bridge, so this is a strong selection bias rather than a strict selection rule — the expected signature of a coupled system in which $\mathcal{L}_\perp$ non-trivially mixes sectors.

Broad-spectrum scan. Extending the scan to $0.05 \le \omega/\omega_c \le 5$ on the same grid reveals four principal peaks, present in both drive spectra but reordered by drive sector:

Peak ω/ω_c	Rung	Predicted	Rel. error
0.1944	3/2	0.207	6.1%
0.2769	2	0.276	0.3%
1.1019	8	1.104	0.2%
4.0513	29.5	4.071	0.5%

In this reduced basis the pion-side and higher target windows resolve to sub-percent proximity. Later controls show that this proximity is not a derived ladder: the half-integer/integer classifier is dense at high frequency, the pion rung is absent in the Path B basis-enrichment test, and the muon-side branch is not stable under either basis enrichment or direct $m_\varphi = \pm 1$ discretisation.

E.4 Box-size dependence and residual uncertainty

The lowest rung is strongly sensitive to the meridional box size, which is consistent with its ring-scale character: core-localised modes are represented accurately by any box that captures the core, while ring-scale Kelvin modes have transverse extent reaching toward $R_e = \xi/\alpha \approx 137\xi$, much larger than any meridional box achievable with the present numerics.

At fixed $n = 41$, varying the meridional half-width L_{box}:

L_{box}/ξ	ω_μ/ω_c (core-drive peak)	Rel. error from 0.207
5	0.215	+3.9%
6	0.200	−3.4%

In the reduced helicity basis the muon-side peak brackets the observed value as the box grows. This was the original motivation for the closure programme, but it is not sufficient evidence for convergence: standard Galerkin enrichment moves or removes the branch, and the direct phi-sector solve finds no corresponding low direct-sector mode.

E.5 Interpretation and comparison to experimental precision

The combined evidence from this reduced calculation suggested:

- a possible muon identification with the lowest core-breathing $\oplus$ Kelvin hybrid coupled through $\mathcal{L}_\perp$, not merely a numerical coincidence.
- driven-response selection rules appeared to support the identification by symmetry: core drive $\leftrightarrow$ muon rung, Kelvin drive $\leftrightarrow$ pion rung.

- the eigenfrequency bracketed the observed value $\omega_\mu/\omega_c \in [0.19, 0.23]$ with residual uncertainty that was initially attributed to finite-meridional-domain effects.

The later closure programme changed this interpretation. Path B showed that the branch is not basis-robust, issue #73 showed that it is not reproduced by a direct $m_\varphi = \pm 1$ grid sector, and issue #76 showed that the current scalar $\mathcal{L} + \mathcal{L}_\perp$ operator does not supply the missing half-integer Berry holonomy. The reduced-basis response remains a diagnostic, not a derivation of $m_{\mu^\pm}$.